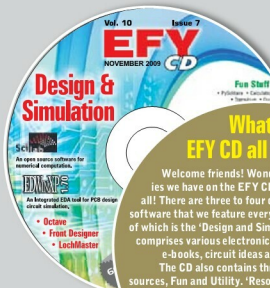


Design & Simulation Bonanza for Engineers

This month we have open source numerical computation software under 'Design and Simulation'; an ebook on how to begin Linux Device Driver programming; Integrated Development Environment software under 'Utilities' and some exciting open source games for the fun lovers.



What's EFY CD all about?

Welcome friends! Wondering what goodies we have on the EFY CD? Allow us to reveal all! There are three to four different categories of software that we feature every month, most important of which is the 'Design and Simulation' category, which comprises various electronics software, datasheets, e-books, circuit ideas and EFY software.

The CD also contains three other sections: Resources, Fun and Utility. 'Resources' contains essential software like Acrobat Reader, Enzip and a few others, which form the basic requirements of any PC.

The 'Fun' segment includes computer games, whereas in 'Utilities' you will find multimedia and IDE software, Internet applications, etc.

Front Designer (Design and Simulation, 5 MB)

Want to ensure your self-made projects look good and presentable? Then you can use Front Designer to create good-looking front panels for your devices. You can comfortably draw functions

for rectangles, polygons, ellipses, labels, drilling, etc. The software also gives you specialised features for rotation, stretching, mirroring, drilling and milling. You can find all these functions in the dockable tool options.

It has the following features: scale-assistant, symbol-library, measures, panel cuts, printing and graphic exports. Here, you can export the front panel into BMP, JPG or EMF and the resolution can be up to 600 dpi.

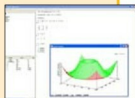
Front Designer is also able to make a printout over several pages. So you can create printouts of up to 180 m. The printout will be divided into several pages automatically. But the software that is provided here in the CD is a demo version, hence you cannot save, print and export the design that you create. For more information visit <http://www.abacom-online.de/uk/html/frontdesigner.html>



Octave (Design and Simulation, 43.6 MB)

Octave is a high-level open source language that has been released under the GNU GPL licence and runs on multiple platforms (Windows, Linux, Solaris, Mac OS). It is written in C++ language using the STL library. You can use Octave for numerical computation and it is compatible with the MATLAB software from MathWorks. While using the software, you will find a command line interface, using which you can solve linear and non-linear problems. The language used here is very similar to MATLAB. Octave is a structured programming language which is similar to C and hence supports many commonly used C library functions. Moreover, it also supports the system calls and functions of the UNIX system. The syntax it uses is matrix-based and hence it supports many matrix functions. The latest stable version of the software is 3.2.3, released on September 21, 2009.

Certain good features of Octave are command and variable name completion that helps you complete of frequently run commands. It has a command history and Boolean operators, increment and decrement operators, variable-length argument lists and much more. Features that Octave shares with MATLAB are matrices as datatypes and support for complex numbers. For more information, visit <http://www.gnu.org/software/octave/>



Scilab

(Design and Simulation, 86 MB)

Scilab is an open source numerical computing software and is licensed under the GPL-compatible CeCILL licence. It was developed by researchers of INRAI in 1990. INRAI created a consortium on May 2003 which maintains Scilab. Being a numerical oriented programming language, Scilab's main data types are matrices. One of the advantages of using matrices as data types is that, many numerical problems can be expressed in a reduced number of code lines as compared to similar solutions using languages such as C, C++ or Fortran. The matrix-based computation also helps you in dynamic typing, automatic memory, etc. Moreover, it can be used for signal processing, statistical analysis, image enhancement, fluid dynamics simulations and numerical optimisation. The software includes a package called Scicos for modelling and simulation of explicit and implicit dynamic systems, including both dynamic, continuous and discrete sub-systems.

It supports multiple operating systems, that is, Linux, Unix and Windows. For more information visit www.scilab.org

